

Objective function

Gradient and descent

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Gradient and descent

The gradient provides useful information about the geometry of the function. In particular, it allows to find directions of descents.

The main argument: Taylor's theorem.

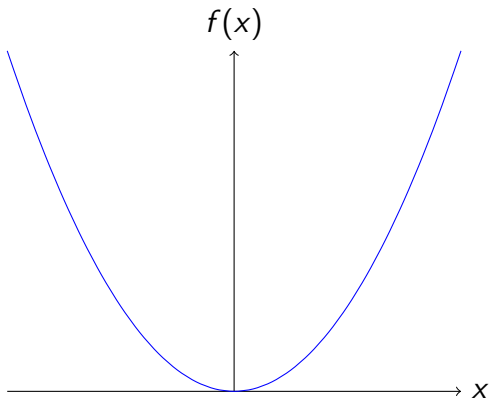
Descent direction

Theorem 2.11

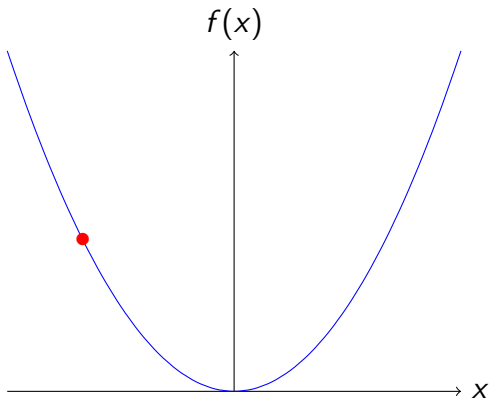
$\exists \eta_1 > 0$ such that

$$f(x + \alpha d) < f(x), \quad \forall 0 < \alpha \leq \eta_1.$$

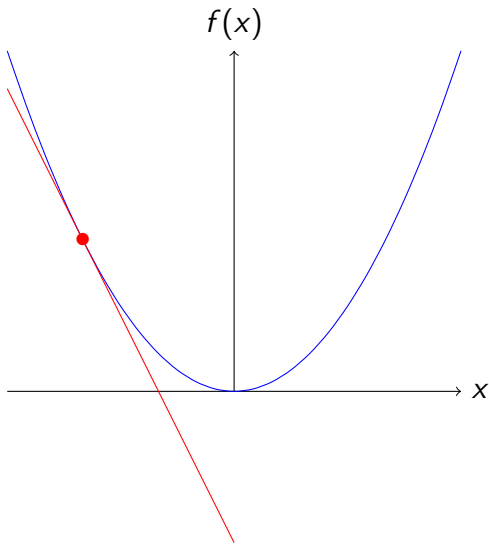
Descent direction



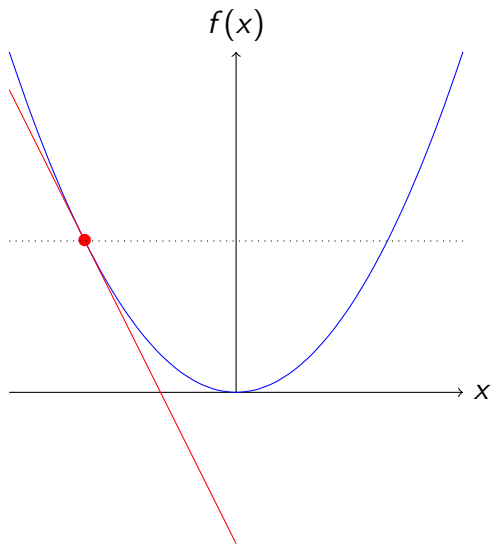
Descent direction



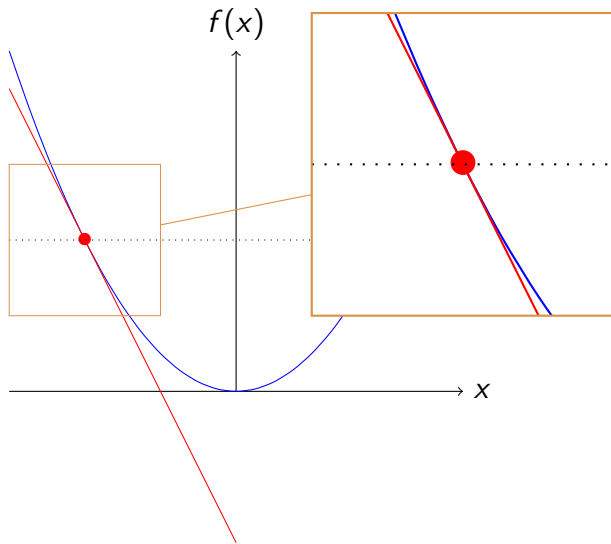
Descent direction



Descent direction



Descent direction

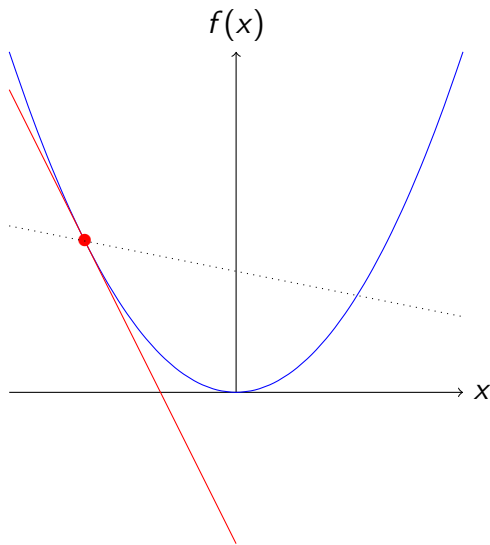


Descent direction

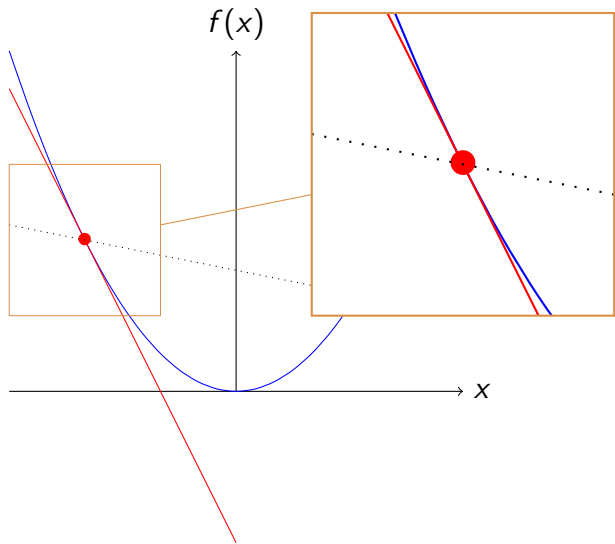
For $\beta < 1$, $\exists \eta_2 > 0$ such that

$$f(x + \alpha d) < f(x) + \alpha \beta \nabla f(x)^T d, \quad \forall 0 < \alpha \leq \eta_2.$$

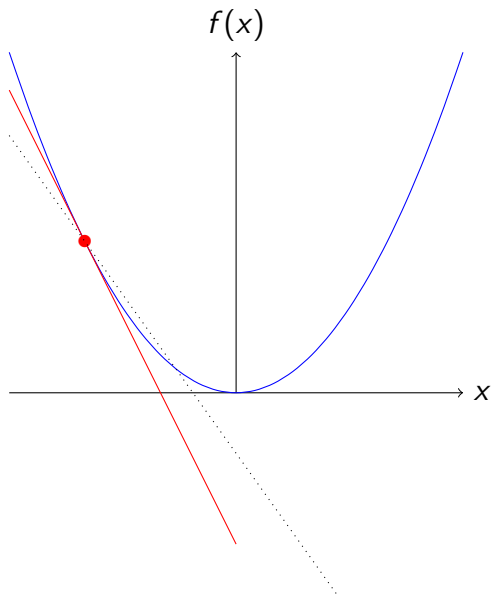
Descent direction: $\beta = 0.1$



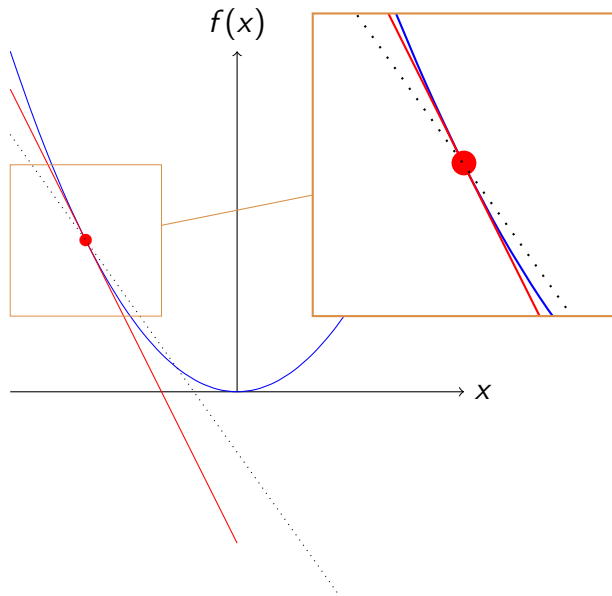
Descent direction: $\beta = 0.1$



Descent direction: $\beta = 0.7$



Descent direction: $\beta = 0.7$



Steepest descent

Theorem 2.12 — Corollary 2.13

Consider

$$d = -\nabla f(x).$$

If $\nabla f(x) \neq 0$, it is a descent direction:

$$d^T \nabla f(x) = -\nabla f(x)^T \nabla f(x) = -\|\nabla f(x)\|^2 < 0.$$

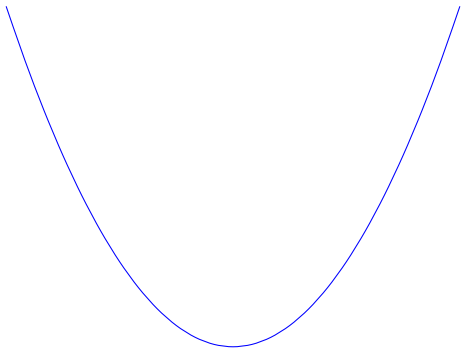
Gradient and convexity

Theorem 2.16: f is convex if and only if

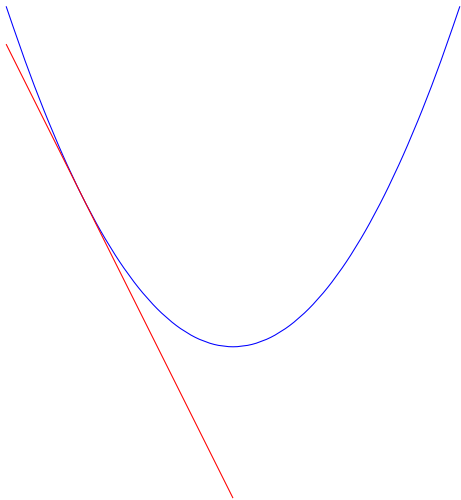
$$f(y) \geq f(x) + (y - x)^T \nabla f(x).$$

The graph of the function is never below the hyperplane tangent.

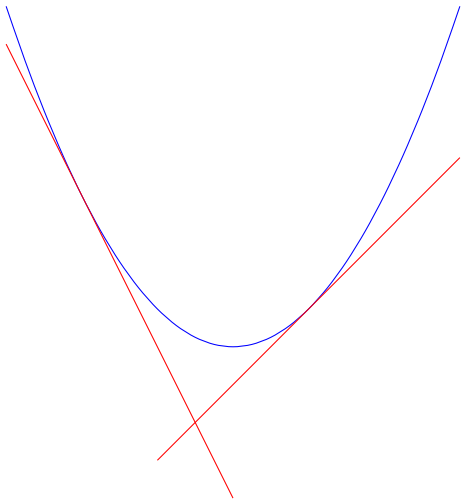
Gradient and convexity



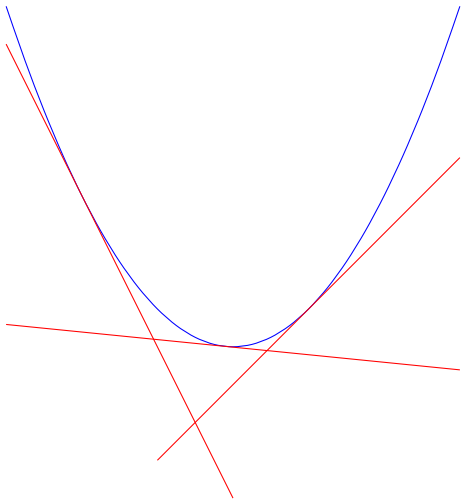
Gradient and convexity



Gradient and convexity



Gradient and convexity



Summary

*The gradient allows to analyze the function along a descent direction.
It quantifies how much it decreases.
It also provides a descent direction (the steepest).
It characterizes the convexity.*